Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU
Department of Computer Science and Engineering

Course Title	Database Management Systems	**Course Code**	BCSPCC403
Course Instructor	Dr. Gururaj T, Prof.Hemashree ,Prof.Shwetha G,. Prof.Roopa & Prof. Santhosh P	**Semester-Section**	IV
IA Test No.	II	**Max. Marks**	30
Date	11-06-2026	**Time**	9.30 to 10.45

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Conceptualize data modelling concepts to draw ER-diagrams for the given scenarios.
CO2	Create efficient queries to retrieve data from the database considered.
CO3	Apply the database normalization concepts to devise efficient databases for the given real time problems.
CO4	Analyse the importance of Transaction processing and concurrency control in efficient database management.
CO5	Differentiate between DBMS and NoSql.

Note: Answer any one full question from each Part.

Q. No.		Question	Marks	RBT Level	CO
		Part A			
1	A	**Consider** the schema for Employee Database: EMP(EmpID, EmpName, Salary, DeptID) DEPT(DeptID, DeptName, ManagerID) **Write** SQL queries to: i. Retrieve the employee name and department name of all employees. ii. List the names of employees working in the "CSE" department. iii. Display employee names along with their salaries and department names.	6	L3	3
		OR			
2	A	**Consider** the schema for Students Database: STUDENT(USN, SName, Sem, DeptID) DEPARTMENT(DeptID, DeptName, HOD) COURSE(CourseID, CourseName, DeptID) **Write** SQL queries to: i. Display student names along with their semester and department names. ii. List all students and the courses offered by their department. iii. Display student names along with the HOD of their department.	6	L3	3
		Part B			
3	A	**Define** Join Dependency and Fifth Normal Form (5NF). **Illustrate** with an example.	6	L2	4
	B	**What** is transaction processing? **Explain** the key transaction and system concepts, such as recovery and concurrency control.	6	L2	4
		OR			
4	A	**What** is Boyce-Codd Normal Form (BCNF)? **Differentiate** it from 3NF with a suitable example where a relation is in 3NF but not in BCNF.	6	L2	4
	B	**Describe** transaction support in SQL, including commands like COMMIT, ROLLBACK and SAVEPOINT.	6	L2	4
		Part C			
5	A	**Discuss** the concept of the Two–Phase Locking (2PL) protocol used in database concurrency control.	6	L2	5
	B	**Explain** the primary CRUD (Create, Read, Update, Delete) operations in MongoDB with suitable examples.	6	L2	5

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		Note: Answer any one full question from each Part.			
Q. No.		**Question**	**Marks**	**RBT Level**	**CO**
		OR			
6	A	**Explain** the concepts of deadlock and starvation in database systems.	6	L2	5
	B	**Describe** the CAP theorem in distributed database systems. **Which** of the three properties are most important in NOSQL systems?	6	L2	5

RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain					
L1 : Remembering	L2 : Understanding	L3 : Applying	L4 : Analysing	L5 : Evaluating	L6 : Creating

Course Instructor **Course Coordinator** **DAAC** **Program Coordinator / HOD**

Bapuji Educational Association (Regd.)

Bapuji Institute of Engineering and Technology, Davanagere - 04

Department of Computer Science and Engineering (CS and allied Branches)

Course Title & Code:	Database Management Systems [BCSPCC403]	Test No.:	2
Semester:	4th Sem	Date:	11-06-2026
Course Instructor	Dr. Gururaj T, Prof. Hemashree, Prog. Shwetha G, Prof. Roopa, Prof. Santhosha P	Marks	30

Qn.	Scheme	Marks
1 A. =	[PART-A] i. Select E.Empname, D.DeptName From EMP E Join Dept D on E.DeptID = D.DeptId; → (2M) iii) Select E.Empname From Emp E Join Dept D on E.DeptID = D.DeptID Where D.DeptName = 'CSE'; → (2M) iii) Select E.EmpName, E.Salary, D.DeptName From Emp E Join Dept D on E.DeptID = D.DeptID. → (2M) Total 6M	

Qn.	Scheme	Marks
	[OR]	
2A	i) Select S.SName, S.Sem, D.DeptName From Student S Join Department D ON S.DeptID = D.DeptID;	→ 2M
	ii) Select S.Sname, C.CourseName From Student S Join Course C on S.DeptID = C.DeptID;	— 2M
	iii) Select S.SName, D.HOD From Student S Join Department D on S.DeptID = D.DeptID	— 2M
		Total (6M)

Qn.	Scheme	Marks

3 a

[PART-B]

Join Dependency:- It is denoted by JD ($R_1, R_2 \ldots R_n$), specified on Relation schema R, specifies a constraints on the states r of R. The constraint states that every legal state r of R should have a nonadditive join decomposition into $R_1, R_2 \ldots R_n$. Hence, for every such r we have

* $(\Pi_{R_1}(r), \Pi_{R_2}(r), \ldots \Pi_{R_n}(r)) = r$.

JD (R_1, R_2) $= (R_1 \cap R_2) \twoheadrightarrow (R_1 - R_2)$

3M

Fifth Normal Form (5NF)

A relation schema R is in 5NF (Project Join normal Form) with respect to a set of F, functional, multivalued, and join Dependencies if, for every non-trivial join Dependency JD ($R_1, R_2 .. R_n$) in F^+, for every R_i is a superkey of R.

3M

Ex:- Relation (Faculty)

Subject	faculty	year
C	Sai	1
DBMS	Ajith	3

(Sub_fac)

Subject	faculty
C	Sai
DBMS	Ajith

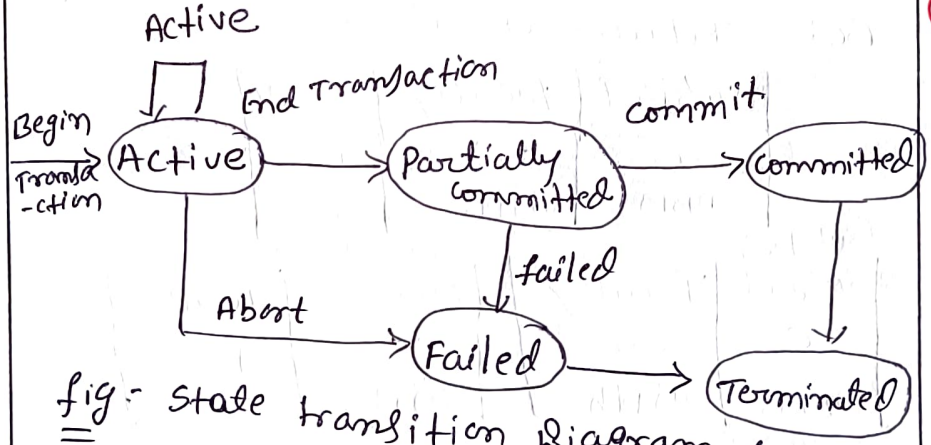
(Fac_year)

faculty	year.
Sai	1
Ajith	3

(Sub_year)

Subject	year
C	1
DBMS	3

Total (6M)

Qn.	Scheme	Marks
3b	**Transaction processing :-** Transaction processing are systems with large databases and hundreds of concurrent users executing database transaction. **Key transaction and system concepts** *) BEGIN_TRANSACTION *) READ & WRITE *) END_TRANSACTION } Explain. *) COMMIT_TRANSACTION *) ROLLBACK (or ABORT) **System Logs.** *) [start_transaction, T] *) [write_item, T, X, old_value, new_value] *) [read_item, T, X] *) [commit, T] *) [abort, T]  fig - State transition Diagram for transaction execution	6M

Qn.	Scheme	Marks
4A	[OR] Boyce-codd Normal Form (BCNF)	

*BCNF is an extension to the 3NF, It is also known as 3.5 NF.

→ A relation schema R is in BCNF if whenever a non trivial functional dependency X → A holds in R, then X is a super key of R.

Example

FD1 : {Student, course} → Instructor

FD2 : Instructor → course.

Solution

S.K = (Student, course, Instructor)$^+$ = {Student, course, Instructor}

C.K (Student, course)$^+$ = {Student, course, Instructor} (course — R.H.S of FD2.)

(Student, Instructor}$^+$ = {Student, Instructor, course}

6M

Student → A

course — B

Instructor - C

R

A	B	C

Table is in 3NF but not in BCNF.

Decomposition of Relation R to BCNF

R1 (Student, Instructor) & R2 (Student, course)

R1 (course, Instructor) & R2 (course, Student)

R1 (Instructor, course) & R2 (Instructor, Student)

Qn.	Scheme	Marks
4b	**Transaction support in SQL** Transaction is a group of SQL operations Executed as a single unit of work, it Ensures data integrity by following Principles of ACID (Automicity, Consisten -cy, Isolation, Durability) **Transaction Commands** *) **Commit** :- Save all changes made During the transaction perm -anently to the database. Ex:- COMMIT; *) **ROLLBACK** :- undoes all changes made in the current transaction Since the last COMMIT. Ex:- ROLLBACK; *) **SAVEPOINT** :- creates a point within a transaction to which you can later rollback. Ex:- SAVEPOINT SP1; Ex:- BEGIN Transaction UPDATE Accounts SET Balance = Balance -1000 WHERE Accountno = 101; SAVEPOINT SP1; UPDATE Accounts SET Balance = Balance + 1000 WHERE Accountno = 102; COMMIT;	2M 4M Total 6M

Qn.	Scheme	Marks

[PART-C]

5A **Two Phase Locking**

It is one of the concurrency control technique. It requires both Locks & unlocks.

2PL is devided into 2 Phases

1) growing Phase 2) Shrinking Phase. — 3M

→ growing phase :- Transaction obtained only locks the transaction, but not release the transaction

→ Shrinking phase :- Locks are released but no new locks are required.

These 2 are having lock point.

Example

```
T1      T2
L1(A)
R(A)
A=A+100
W(A)
L1(B)
unlock(A)
 :
 :
```

Variations of 2PL

1) Conservative (Static)
2) Strict 2PL
3) Rigorous 2PL. → 3M

⇒ Conservative 2PL.

Locks all the Data item before transaction. After completion all operation then only release.

⇒ Strict 2PL

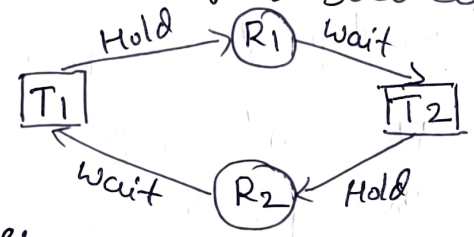
All exclusive locks should hold until commit / Abort.

⇒ Rigorous 2PL

All Shared and exclusive locks should hold until commit / Abort.

Total 6M

Qn.	Scheme	Marks
5B	MongoDB :- Store the data in documents similar to JSON (Java Script object notation) → Document contains pairs of fields & value. CRUD operations in MongoDB C - Create R - Read U - update D - Delete	2M
1)	Create :- use COMPANY → Database creation db.createCollection("Employee")	
2)	Insert Documents. db.Employee.insertOne({Name: 'ABC', Branch: 'AIML'}) db.Employee.insertMany([{Name: 'XYZ', Branch: 'CS'}, {Name: 'ABC'}])	
3)	Read :- Read the documents from Employee collection db.Employee.find().	4M
4)	update db.Employee.updateOne({Name: 'ABC'}, {$set: {Branch: 'IS'}}) db.Employee.updateMany({}, {$set: {Branch: 'IS'}})	Total 6M
5)	Delete :- delete one (or) more documents from Employee collection db.Employee.deleteOne({Name: 'ABC'}) (or) db.Employee.deleteMany({}, {Name: 'ABC'})	

Qn.	Scheme	Marks
6a (OR)	**Deadlock** occurs when two (or) more transactions are waiting indefinitely for each other to release resources.  **Necessary conditions for Deadlock occur** 1) Mutual Exclusion 2) Hold & wait 3) NO-preemption 4) Circular wait **Deadlock Avoidance | prevention** 1) wait and Die technique 2) Wound and wait technique } Explain **Deadlock Detection Technique** 1) wait for Graph 2) Resource Allocation Graph. **Starvation** Starvation occurs when a transaction wait for a resource for an unbounded perior because other transactions ~~can~~ continuously get priority.	Total 6M

Qn.	Scheme	Marks
6b	CAP Theorem states that a distributed database can provide only two of the three properties : consistency, Availability and partition Tolerance. → Most NOSQL databases prioritize Availability and partition Tolerance to ensure continuous service and fault tolerance in distributed system databases. Properties C (Consistency) :- All nodes see the same data at same time. A (Availability) :- Every request receive a response, even if some nodes are fail. PC (Partition Tolerance) :- The system continues to operate despite network communication failure between nodes.	→3M →3M Total 6M

Course coordinators

For DAAC 12/6/24

Program coordinator (HOD. CS & Engg)

BOE Chairman